

When Bonds Stop Hedging Equities

Supply-Shock Regimes and Canadian Portfolio Construction

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Executive Summary

Government bonds are expected to diversify equity risk in Canadian balanced portfolios, but that benefit is not constant. Using a structural VAR, a Markov-switching model, and DCC-GARCH correlations, I find that bond-equity correlation is more than twice as high during persistent supply-shock regimes than during normal regimes — averaging 0.25 versus 0.11. Inflation levels alone provide a much weaker signal. What matters is not how high inflation is, but whether supply shocks have become persistent and volatile.

One natural interpretation is that persistent supply shocks place the Bank of Canada in a difficult trade-off, where tightening to defend the inflation target simultaneously pressures both bond prices and equity valuations. The empirical pattern is consistent with this channel, though the mechanism is not directly tested here.

The portfolio implication is that 60/40 becomes less resilient precisely when the bond hedge is most needed. Among allocations tested, a 50/30/20 split across Canadian equities, Canadian bonds, and unhedged gold delivered the strongest risk-adjusted performance — improving return, reducing volatility, and cutting maximum drawdown. The benefit of unhedged over hedged gold suggests that USD-linked defensive exposure contributed alongside bullion.

Research Question

The project aims to answer: when does the bond-equity correlation rise, what macro regime drives the shift, and how should Canadian multi-asset portfolios adapt?

The core hypothesis is that bond-equity correlation becomes less favorable when supply/inflation shocks are not isolated events but part of a persistent high-pressure regime. In that environment, rate risk, inflation persistence, and weaker growth can pressure both equities and bonds at the same time.

Data and Proxies

The analysis uses public monthly data. Market proxies are investable ETFs where possible. Macro data are Canadian GDP, CPI inflation, and the Bank of Canada policy rate. The SVAR-linked sample begins in 2009 because of aligned macro-data availability.

Role	Proxy	Interpretation
Canadian equities	XIC.TO	Broad S&P/TSX capped composite exposure
Canadian bonds	XBB.TO	Broad Canadian universe bond exposure
Short bonds	XSB.TO	Lower-duration Canadian bond substitute
Gold, unhedged	CGL-C.TO / CGL.C	Gold bullion plus USD/CAD currency exposure
Gold, CAD-hedged	CGL.TO / CGL	Gold bullion with CAD hedge
Commodities	DBC	Broad commodity exposure
Currency proxy	FXC	CAD/USD proxy
Macro	GDP, CPI, BoC rate	Growth, inflation, and policy conditions

Methodology

SVAR Shock Identification

The baseline macro system is:

$$Y_t = [\Delta GDP_t, \pi_t, i_t]$$

where ΔGDP_t is monthly real GDP growth, π_t is year-over-year CPI inflation, and i_t is the Bank of Canada policy rate. A robustness version uses changes in inflation and changes in the policy rate to reduce persistence.

Shock identification uses sign restrictions:

Shock	GDP	Inflation	Policy rate	Interpretation
Demand	+	+	+	Growth and inflation rise together
Supply/inflation	-	+	+	Inflation rises while growth weakens
Monetary policy	-	-	+	Exogenous tightening cools activity

The SVAR identifies the type of macro shock. The Markov model then asks whether supply shocks are isolated or persistent.

Markov Switching on Supply Shocks

A two-state Markov-switching model is estimated on the SVAR supply/inflation shock series:

$$s_t \sim \begin{cases} \text{normal supply state} & \text{small, well-behaved shocks} \\ \text{high-pressure supply state} & \text{larger and more volatile shocks} \end{cases}$$

This is intentionally different from a Markov model on inflation levels. Inflation alone is too blunt: high inflation is not always a supply-shock regime. The relevant portfolio question is whether the supply-shock process itself has entered a persistent high-pressure state.

The regime labels used in this brief are based on smoothed probabilities, which use the full sample and are appropriate for historical diagnosis. A real-time allocation strategy would need filtered probabilities, which use only information available up to each month.

DCC-GARCH Correlation

XIC and XBB monthly returns are modeled using univariate GARCH(1,1) volatility models and a two-asset DCC correlation process. The DCC estimate measures whether bond-equity correlation changes with the supply-shock regime.

Empirical Results

Persistent Supply Shocks and Correlation

The DCC-GARCH result is the cleanest bridge from macro regime to portfolio risk. The historical smoothed-probability result is shown first.

Supply-shock regime	Obs.	Avg. DCC corr.	Median	Min	Max
High-pressure supply	83	0.245	0.263	0.069	0.434
Normal supply	112	0.110	0.097	-0.087	0.421

Bond-equity correlation is more than twice as high in the high-pressure supply state. This supports the central thesis: the 60/40 hedge weakens when inflationary supply shocks become persistent. The result is robust to filtered probabilities, which are closer to a real-time regime signal.

Filtered supply-shock regime	Obs.	Avg. DCC corr.	Median	Min	Max
High-pressure supply	84	0.250	0.281	0.069	0.434
Normal supply	111	0.105	0.096	-0.087	0.416

This addresses the real-time critique: the correlation result is not only visible in hindsight. Even using information available up to each month, high-pressure supply regimes are associated with materially higher bond-equity correlation.

Why Inflation Alone Is Insufficient

A separate Markov model on inflation levels gives a weaker signal. High-inflation months do not produce a materially higher bond-equity correlation than low-inflation months.

Inflation regime	Obs.	Avg. inflation	XIC/XBB corr.	60/40 return	60/40 max DD
High inflation	111	3.23%	0.296	8.8%	-11.5%
Low inflation	91	1.30%	0.330	7.1%	-15.4%

This is an important negative result. Inflation level by itself is not the main portfolio signal. The more relevant signal is the persistence and volatility of supply/inflation shocks.

Portfolio Construction

Portfolio tests compare practical allocations across the Markov supply-shock regimes. The high-pressure supply state is the stress environment of interest.

High-Pressure Supply Regime: Smoothed Historical Diagnosis

The best tested portfolio in the high-pressure supply regime is 50% XIC, 30% XBB, and 20% unhedged gold. It improves return, volatility, maximum drawdown, worst month, and Sharpe ratio versus the 60/40 baseline.

Higher-equity allocations improve return in some cases but worsen drawdowns and downside risk. The result therefore does not argue for simply adding equity. It argues for reducing reliance on duration and adding diversifiers that are not tightly linked to Canadian rates or Canadian equity beta.

Portfolio	Return	Vol.	Max DD	Worst month	Sharpe
60/40 XIC/XBB	5.0%	10.8%	-15.4%	-12.5%	0.30
70/30 XIC/XBB	5.5%	11.9%	-17.4%	-14.0%	0.32
75/15/10 unhedged gold	7.6%	12.4%	-17.6%	-13.9%	0.49
60/30/10 unhedged gold	6.8%	10.7%	-14.5%	-11.7%	0.48
50/30/20 unhedged gold	8.1%	9.6%	-11.6%	-9.3%	0.68
50/30/20 hedged gold	7.6%	10.2%	-12.9%	-10.4%	0.58
45/25/15/10/5 real-asset mix, unhedged	6.1%	8.8%	-12.0%	-9.6%	0.49
60/25/10/5 commodities/CAD	4.8%	11.3%	-18.9%	-13.9%	0.26

The poor performance of the commodities/CAD portfolio is also informative. A broad commodity index such as DBC is not the same as a targeted Canadian energy hedge: it includes energy, agricultural commodities, and metals, whose returns may not line up cleanly with Canadian supply-shock stress. In this sample, that broad commodity/CAD sleeve added volatility and drawdown risk rather than improving diversification.

Filtered Real-Time-Style Portfolio Check

Using filtered regime probabilities, the portfolio results remain supportive. The filtered high-pressure supply state contains more months in which Canadian equities performed well, so higher-equity portfolios show stronger returns. Even so, the best risk-adjusted result remains the 50/30/20 allocation with unhedged gold.

Portfolio	Return	Vol.	Max DD	5% VaR	Sharpe
60/40 XIC/XBB	8.5%	9.4%	-11.1%	-3.6%	0.66
75/15/10 unhedged gold	11.2%	10.8%	-12.4%	-4.0%	0.87
70/20/10 unhedged gold	10.8%	10.4%	-11.9%	-3.8%	0.86
50/30/20 unhedged gold	10.4%	8.7%	-10.0%	-2.6%	1.01
50/25/15/10 short/gold, unhedged	9.0%	8.2%	-9.6%	-2.8%	0.89
45/25/15/10/5 real-asset mix, unhedged	8.4%	7.8%	-9.2%	-2.7%	0.84

The filtered results strengthen the portfolio conclusion. A higher-equity allocation can raise return, but the 50/30/20 unhedged-gold mix provides the more attractive balance of return, volatility, drawdown, and left-tail risk.

Hedged vs. Unhedged Gold

Unhedged gold outperforms CAD-hedged gold across the most relevant comparisons.

Portfolio	Return	Vol.	Max DD	Sharpe
50/30/20 unhedged gold	8.1%	9.6%	-11.6%	0.68
50/30/20 hedged gold	7.6%	10.2%	-12.9%	0.58
60/30/10 unhedged gold	6.8%	10.7%	-14.5%	0.48
60/30/10 hedged gold	6.6%	11.1%	-15.2%	0.44

This means the gold result is not purely a bullion result. For Canadian investors, unhedged gold combines gold exposure with USD-linked defensive currency exposure. The hedge may reflect gold, USD strength,

CAD weakness, or all three. That distinction matters: the result should be interpreted as support for non-CAD, non-duration diversification, not as a simple claim that gold alone hedges Canadian inflation.

Investment Implications

The evidence points to four practical implications.

First, 60/40 is regime-dependent. Bonds still play a useful role in normal regimes, but their diversification benefit weakens when supply/inflation shocks become persistent.

Second, inflation level alone is not enough. A high/low inflation regime does not cleanly identify the portfolio risk state. The source and persistence of inflation pressure matter more.

Third, duration substitution should be selective. Short bonds modestly reduce volatility, but do not fully solve the problem. Broad commodity/CAD exposure performed poorly in the tested high-pressure supply state, suggesting that commodity hedges need to be targeted rather than assumed.

Fourth, unhedged gold was the most effective tested diversifier. The best result came from a meaningful unhedged gold allocation, suggesting that USD-linked defensive exposure contributed to the hedge. CAD-hedged gold helped less, which points to currency exposure as part of the diversification mechanism.

Limitations

The analysis uses public ETF proxies and a relatively short macro-aligned sample beginning in 2009. Model results depend on the SVAR identification scheme, the Markov specification, and the ETF histories available. The portfolio tests are historical and should not be interpreted as optimized recommendations. Smoothed probabilities are useful for historical diagnosis, while filtered probabilities are more relevant for real-time implementation; both are reported, but a true trading strategy would require a fully out-of-sample design. Gold's benefit is also partly currency-related, so it should not be described as a pure inflation hedge.

Conclusion

The main finding is that Canadian bond-equity diversification weakens not simply when inflation is high, but when supply/inflation shocks become persistent. In those regimes, DCC-GARCH correlation between Canadian equities and Canadian bonds is materially higher, and the traditional 60/40 allocation becomes less resilient.

The mechanism is a policy credibility and reaction-function problem. Persistent supply shocks put the Bank of Canada in a difficult tradeoff: tightening is needed to defend the inflation target, but tighter policy can also weaken growth. That policy trilemma can create simultaneous pressure on bonds through higher rates and on equities through weaker earnings, margins, and discount rates.

For Canadian multi-asset portfolios, the practical response is not to abandon bonds or mechanically add commodities. The evidence supports a more measured approach: reduce reliance on duration, consider lower-duration substitutes, and add diversifiers that can perform when Canadian rates and equities are pressured together. In this sample, unhedged gold was the strongest tested complement, likely because it combined bullion exposure with USD-linked defensive currency exposure.